

ME 333
MECHANICS OF MATERIALS

Lecture 3: Generalized Strain Description II

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Generalized Strain Description

Steps :

- 1 Consider a point $P(x, y, z)$ in an unloaded state (undeformed and unstressed)
- 2 Upon loading point P moves to $P'(x', y', z')$
- 3 The primed coordinates are functions of unprimed coordinates and vice-versa, i.e.,

$$x' = x'(x, y, z); \quad x = x(x', y', z')$$

$$y' = y'(x, y, z); \quad y = y(x', y', z')$$

$$z' = z'(x, y, z); \quad z = z(x', y', z')$$

Requirement :

- 1 Above functions are continuous and differentiable
- 2 Above functions are invertible and single valued

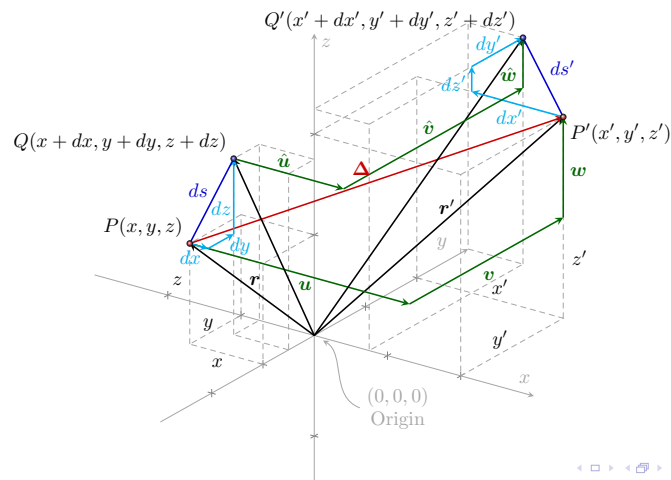
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Generalized Strain Description

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Generalized Strain Description

Generalized Strain Description

Objective : Define strain at point P in terms of (u, v, w) and (x, y, z)

- Step 1 : The *engineering strain measure* can be defined in a fully three dimensional setting as:

$$\epsilon_E = \left(\frac{ds' - ds}{ds} \right); \quad \epsilon_E > -1$$

- Step 2 : The differential lengths ds and ds' can be defined as:

$$ds'^2 = dx'^2 + dy'^2 + dz'^2$$

$$ds^2 = dx^2 + dy^2 + dz^2$$

Using the relation between primed and unprimed spatial coordinates:

$$x' = x + u; \quad y' = y + v; \quad z' = z + w$$

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Generalized Strain Description

- **Step 3 :** Relating the differential forms of primed and unprimed quantities:

$$\begin{aligned} dx' &= \frac{\partial x'}{\partial x} dx + \frac{\partial x'}{\partial y} dy + \frac{\partial x'}{\partial z} dz = \left(1 + \frac{\partial u}{\partial x}\right) dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz \\ dy' &= \frac{\partial y'}{\partial x} dx + \frac{\partial y'}{\partial y} dy + \frac{\partial y'}{\partial z} dz = \frac{\partial v}{\partial x} dx + \left(1 + \frac{\partial v}{\partial y}\right) dy + \frac{\partial v}{\partial z} dz \\ dz' &= \frac{\partial z'}{\partial x} dx + \frac{\partial z'}{\partial y} dy + \frac{\partial z'}{\partial z} dz = \frac{\partial w}{\partial x} dx + \frac{\partial w}{\partial y} dy + \left(1 + \frac{\partial w}{\partial z}\right) dz \end{aligned}$$

(or)

$$\begin{Bmatrix} dx' \\ dy' \\ dz' \end{Bmatrix} = \begin{bmatrix} \left(1 + \frac{\partial u}{\partial x}\right) & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \left(1 + \frac{\partial v}{\partial y}\right) & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \left(1 + \frac{\partial w}{\partial z}\right) \end{bmatrix} \begin{Bmatrix} dx \\ dy \\ dz \end{Bmatrix}$$

Generalized Strain Description

- Dividing by ds on both sides :

$$\begin{Bmatrix} \frac{dx'}{ds} \\ \frac{dy'}{ds} \\ \frac{dz'}{ds} \end{Bmatrix} = \begin{bmatrix} \left(1 + \frac{\partial u}{\partial x}\right) & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \left(1 + \frac{\partial v}{\partial y}\right) & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \left(1 + \frac{\partial w}{\partial z}\right) \end{bmatrix} \begin{Bmatrix} \frac{dx}{ds} \\ \frac{dy}{ds} \\ \frac{dz}{ds} \end{Bmatrix}$$

$$= \begin{bmatrix} \left(1 + \frac{\partial u}{\partial x}\right) & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \left(1 + \frac{\partial v}{\partial y}\right) & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \left(1 + \frac{\partial w}{\partial z}\right) \end{bmatrix} \begin{Bmatrix} l \\ m \\ n \end{Bmatrix}$$

Generalized Strain Description

- **Step 4 :** Squaring the engineering strain measure in Step 1:

$$\begin{aligned} \epsilon_E &= \left(\frac{ds' - ds}{ds}\right) = \frac{ds'}{ds} - 1 \\ \epsilon_E^2 &= \left(\frac{ds'}{ds} - 1\right)^2 = \left(\frac{ds'}{ds}\right)^2 - 2\frac{ds'}{ds} + 1 \end{aligned}$$

- **Step 5 :** Eliminating the linear term (ds'/ds) from above equation :

$$\begin{aligned} \frac{1}{2}\epsilon_E^2 + \epsilon_E &= \frac{1}{2} \left[\left(\frac{ds'}{ds}\right)^2 - 1 \right] = M \\ &= \frac{1}{2} \left[\left(\frac{dx'}{ds}\right)^2 + \left(\frac{dy'}{ds}\right)^2 + \left(\frac{dz'}{ds}\right)^2 - 1 \right] \end{aligned}$$

Generalized Strain Description

- **Step 6 :** Using Step 3 in Step 5 (substituting and simplifying):

$$\begin{aligned} \frac{1}{2}\epsilon_E^2 + \epsilon_E &= l^2 \left(\frac{\partial u}{\partial x} + \frac{1}{2} \left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 + \left(\frac{\partial w}{\partial x}\right)^2 \right] \right) + \\ &+ m^2 \left(\frac{\partial v}{\partial y} + \frac{1}{2} \left[\left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2 + \left(\frac{\partial w}{\partial y}\right)^2 \right] \right) + \\ &+ n^2 \left(\frac{\partial w}{\partial z} + \frac{1}{2} \left[\left(\frac{\partial u}{\partial z}\right)^2 + \left(\frac{\partial v}{\partial z}\right)^2 + \left(\frac{\partial w}{\partial z}\right)^2 \right] \right) + \\ &+ lm \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} + \frac{\partial u}{\partial x} \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \frac{\partial v}{\partial y} + \frac{\partial w}{\partial x} \frac{\partial w}{\partial y} \right) + \\ &+ mn \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} + \frac{\partial u}{\partial y} \frac{\partial u}{\partial z} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \frac{\partial w}{\partial z} \right) + \\ &+ nl \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} + \frac{\partial u}{\partial x} \frac{\partial u}{\partial z} + \frac{\partial v}{\partial x} \frac{\partial v}{\partial z} + \frac{\partial w}{\partial x} \frac{\partial w}{\partial z} \right) \end{aligned}$$

Generalized Strain Description

- The term on the LHS is defined as *magnification factor* denoted by M and the RHS can be re-written in a simpler form as:

$$M = \frac{1}{2} \epsilon_E^2 + \epsilon_E = l^2 \epsilon_{xx} + m^2 \epsilon_{yy} + n^2 \epsilon_{zz} + 2lm \epsilon_{xy} + 2ln \epsilon_{xz} + 2mn \epsilon_{yz}$$

where: $(\epsilon_{xx}, \epsilon_{yy}, \epsilon_{zz})$ are the *normal strains* and $(\epsilon_{xy}, \epsilon_{yz}, \epsilon_{zx})$ are the *shear strains*. The strain, therefore, can be represented as:

$$\epsilon = \begin{bmatrix} \epsilon_{xx} & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{yx} & \epsilon_{yy} & \epsilon_{yz} \\ \epsilon_{zx} & \epsilon_{zy} & \epsilon_{zz} \end{bmatrix}$$

- Special Case :** If line element is parallel to x - axis ($l = 1, m = 0, n = 0$):

$$M = M_x = \epsilon_{xx}$$

Generalized Strain Description

- The components of strain at point P in terms of (u, v, w) and (x, y, z) are:

$$\epsilon_{xx} = \frac{\partial u}{\partial x} + \frac{1}{2} \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial w}{\partial x} \right)^2 \right]$$

$$\epsilon_{yy} = \frac{\partial v}{\partial y} + \frac{1}{2} \left[\left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial y} \right)^2 \right]$$

$$\epsilon_{zz} = \frac{\partial w}{\partial z} + \frac{1}{2} \left[\left(\frac{\partial u}{\partial z} \right)^2 + \left(\frac{\partial v}{\partial z} \right)^2 + \left(\frac{\partial w}{\partial z} \right)^2 \right]$$

$$\epsilon_{xy} = \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} + \frac{\partial u}{\partial x} \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \frac{\partial v}{\partial y} + \frac{\partial w}{\partial x} \frac{\partial w}{\partial y} \right)$$

$$\epsilon_{yz} = \frac{1}{2} \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} + \frac{\partial u}{\partial y} \frac{\partial u}{\partial z} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \frac{\partial w}{\partial z} \right)$$

$$\epsilon_{zx} = \frac{1}{2} \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} + \frac{\partial u}{\partial x} \frac{\partial u}{\partial z} + \frac{\partial v}{\partial x} \frac{\partial v}{\partial z} + \frac{\partial w}{\partial x} \frac{\partial w}{\partial z} \right)$$

Generalized Strain Description : Summary

Objective : Define strain at point P in terms of (u, v, w) and (x, y, z)

- Step 1 :** Consider the *engineering strain* in 3D in terms of ds and ds' .
- Step 2 :** Define the differential lengths ds and ds' in terms of their components (dx, dy, dz) and (dx', dy', dz') , respectively.
- Step 3 :** Relate the differential forms of primed and unprimed quantities and further express in terms of direction cosines.
- Steps 4 & 5 :** Squaring the engineering strains and eliminating the linear term (ds'/ds) from the resulting equation.
- Step 6 :** Collecting the coefficients of the squared and coupled direction cosines to write down the *magnification factor* M .

Strain Tensor

- The strains are components of a *second order symmetric tensor* and can be represented as:

$$\epsilon = \begin{bmatrix} \epsilon_{xx} & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{yx} & \epsilon_{yy} & \epsilon_{yz} \\ \epsilon_{zx} & \epsilon_{zy} & \epsilon_{zz} \end{bmatrix}$$

Here the diagonal terms result in *dilation* or volume change while the off-diagonal terms cause *distortion* or shape changes and rotation of material fibers.

Q How and why did we conclude that the coefficients that go along with the squared and coupled direction cosine terms are indeed components of a second order tensor ?

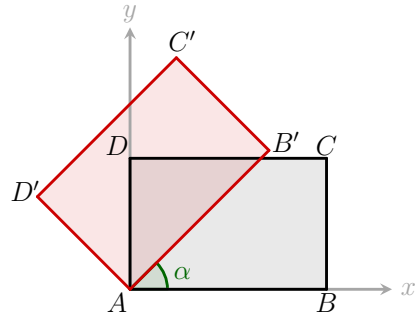
Examples

[Example 7] Rigid rotation about the z-axis: (Repeated)

$$\begin{aligned} x' &= x + u; & y' &= y + v; & z' &= z + w \\ &= x \cos \alpha + y \sin \alpha; & &= -x \sin \alpha + y \cos \alpha; & &= z \end{aligned}$$

- Using the derived strain measures :

$$\begin{aligned} \epsilon_{xx} &= \left(\frac{\partial u}{\partial x} + \frac{1}{2} \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 \right] \right) \\ &= (\cos \alpha - 1) + \frac{1}{2} [(\cos \alpha - 1)^2 + (-\sin \alpha)^2] \\ &= 0 \end{aligned}$$



Examples

[Example 8] Derive the strains using small and large strain theories. Given:

$$x' = 0.99x; \quad y' = 0.99y; \quad z' = 0.99z$$

- Knowing that :

$$x' = x + u; \quad y' = y + v; \quad z' = z + w$$

- The displacement fields are:

$$u = -0.01x; \quad v = -0.01y; \quad w = -0.01z$$

- Based on small & large strain theory:

$$\begin{aligned} \epsilon_{xx} &= -0.01; & \epsilon_{yy} &= -0.01; & \epsilon_{zz} &= -0.01 \\ M_x &= -0.01; & M_y &= -0.01; & M_z &= -0.01 \end{aligned}$$

Examples

[Example 9] Derive the strains using small and large strain theories. Given:

$$x' = 2x; \quad y' = 2y; \quad z' = 2z$$

- Knowing that :

$$x' = x + u; \quad y' = y + v; \quad z' = z + w$$

- The displacement fields are:

$$u = x; \quad v = y; \quad w = z$$

- Based on small & large strain theory:

$$\begin{aligned} \epsilon_{xx} &= 1; & \epsilon_{yy} &= 1; & \epsilon_{zz} &= 1 \\ M_x &= \frac{3}{2}; & M_y &= \frac{3}{2}; & M_z &= \frac{3}{2} \end{aligned}$$

Strain Measure

Q For a given Strain field is the displacement field unique?

Reading Assignment

Chapter 2: Section 2.7.1